# Infrared Attention Network for Woodland Segmentation Using Multispectral Satellite Images

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***Abstract*— Semantic segmentation of the remote sensing images (RSIs) has attracted increasing interest in recent years. However, large-area segmentation of the woodland presents challenges. The wide distribution and diverse tree species of the woodland make feature extraction difficult. For this reason, an infrared attention network (InfAttNet) is proposed to extract woodland from multispectral RSIs. InfAttNet has an extra infrared spectral encoder which makes use of the sensitivity of vegetation to near-infrared and red edge spectrums. This extra encoder applies learning about vegetation to improve woodland segmentation. Several attention blocks are designed to enhance learning about vegetation features and so improve the performance. In addition, a new dataset is built, containing a large number of woodland RSIs and covering several typical woodland distribution regions in China. The experimental results demonstrate that compared with other networks, InfAttNet has the highest accuracy and is capable of rapid extraction of the woodland in RSIs.**

***Index Terms*— Attention block, deep learning, infrared spectrums, remote sensing image (RSI), woodland segmentation.**

## I. Introduction

TO QUICKLY grasp the overall picture of woodland in forest while balancing data quality and data acquisition difficulty, wide field of view (WFV) remote sensing images (RSIs) are usually considered. WFV data contain large-area Earth observation information, while visible light multispectral imagery provides high resolution, rich color, and spectral information. These properties may help the analysis of woodland distributions. Since the launch of the GaoFen-6 (GF-6) satellite in June 2018, a large number of WFV multispectral RSIs have been generated [1]. These images have eight spectrums covering the red edge frequencies that are useful for vegetation classification and significant for Earth observation.

Compared with other RSIs, the WFV data have large observation area and include more complex land categories and higher processing efficiency requirements. This characteristic of the WFV RSIs makes it more appropriate to use semantic segmentation to perform land classification [2], [3]. For woodland observations, many machine learning methods have been used for woodland extracting from RSIs and have achieved good results [4], [5]. Compared with the traditional machine learning, deep-learning-based methods effectively extract deep characteristics and avoid the cognitive limitations of experts' knowledge [6]. In recent years, researchers have used semantic segmentation methods based on deep learning to extract woodland. The dense fully convolutional network proposed by [7] conducted semantic segmentation of the GaoFen-2 RSIs and extracted woodland in Chenzhou, China. [8] used a convolutional encoder neural network, which was based on a convolutional network, to extract vegetation from GaoFen-2 images. [9] designed a balanced batch structure on U-net to segment woodland. However, some of these methods did not fully consider the characteristics of woodland; some were suitable for small-scale woodland segmentation but less suitable for large areas. Woodland has uneven distributions and various tree species and presents complex conditions for remote sensing imaging. Thus, the classic deep learning model encounters challenges in the task of segmenting woodland from RSIs.

In addition to woodland segmentation, there are many other land-type segmentation tasks. In such tasks, woodland (or called trees) is treated as one of many categories. Deep convolutional neural network (DCNN) was adapted to the semantic segmentation of multispectral RSIs and obtained high classification accuracy in the tree category [2]. [10] presented two segmentation networks: symmetrical normal-shortcut fully convolutional networks and symmetrical dense-shortcut fully convolutional networks on the high-resolution RSIs datasets. [11] developed a Markov random-field-based framework to incorporate the multigranularity information and multilayer semantic information, to obtain a good segmentation performance. [12] designed the swin transformer U-net (ST-UNet), which can enhance the feature representation ability and reduce the loss of detailed information. The primary

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goal of these tasks was to enhance the segmentation results for all the categories. As a category, woodland has a larger area than other categories, such as roads and ponds. This makes the tasks easier to extract features from woodland than others and the segmentation result of woodland is better than others. For this reason, there was no specific optimization strategy for woodland segmentation in these tasks, and instead they preferred to enhance the performance of network, optimized other categories, and improved the overall segmentation results.

To solve these problems, this work uses the visible light multispectral RSIs of the GF-6 satellite and proposes an infrared attention network (InfAttNet) to extract the characteristics of the woodland category using infrared spectrums. The mentioned infrared spectrums contain the near-infrared and red edge spectrums. More specifically, the main contributions are summarized as follows.

1) This article analyzes infrared spectrums exhibited by woodland as vegetation, which is incorporated into the deep learning framework, so as to extract high-dimensional features that are specific to woodland. These features are successfully applied for woodland segmentation on RSIs containing the near-infrared and red edge spectrums.
2) A dual-encoder segmentation network and a series of attention mechanisms (AMs) are proposed to enhance feature extraction. The dual-encoder designed to extract the overall features of an image enhances the learning of vegetation characteristics in the infrared spectrums. The reasonable attention blocks improve the effect of segmentation. The experiments demonstrate that the proposed InfAttNet achieves state-of-the-art performance.
3) A woodland remote sensing dataset is built, covering a large area and containing many tree species. The dataset contains several Chinese cities and is labeled by professionals with large numbers and high accuracy.

The remainder of this article is organized as follows. Section II reviews the technologies on which InfAttNet is based. Section III describes the specific structure of InfAttNet. Section IV provides the proposed dataset and compares InfAttNet with mainstream networks. Section V reports on ablation experiments on InfAttNet to explore the usefulness of each block. Section V draws conclusions and briefly discusses future work.

## II. Related Work

### A. Red Edge Effect of Vegetation

The red edge effect (REE) of vegetation refers to the phenomenon of vegetation undergoing a rapid change in reflectivity in the near-infrared range of the electromagnetic spectrum. REE is caused by two optical properties of plant tissues: high internal leaf scattering leading to large near-infrared reflectivity and chlorophyll absorption causing low reflectivity in the red spectrum. The special reflection features of vegetation to electromagnetic waves result in a rapid change between 680 and 750 nm [13], [14]. REE is used to build vegetation indices of plant status, to monitor and identify the chlorophyll content and biological activity of vegetation. Some vegetation indices based on REE can be used to determine the growth status of plants. In addition, the different reflectance changes in plants are used to distinguish plant types.

TABLE I

Detailed Parameters of GF-6 16-Meter Multispectral Camera. All the Values Represent Wave Lengths (WL, unit$\mu$m) of Spectral. The Infrared Spectrums Contain S4, S5, and S6

| Spectrum Name | WL Range (unit:$\mu$m) | WL Center (unit:$\mu$m) |
|---|---|---|
| S1 (blue) | 0.45-0.52 | 0.478 |
| S2 (green) | 0.52-0.59 | 0.528 |
| S3 (red) | 0.63-0.69 | 0.660 |
| S4 (near infrared) | 0.77-0.89 | 0.806 |
| S5 (red edge 1) | 0.69-0.73 | 0.710 |
| S6 (red edge 2) | 0.73-0.77 | 0.744 |
| S7 (purple) | 0.40-0.45 | 0.442 |
| S8 (yellow) | 0.59-0.63 | 0.596 |

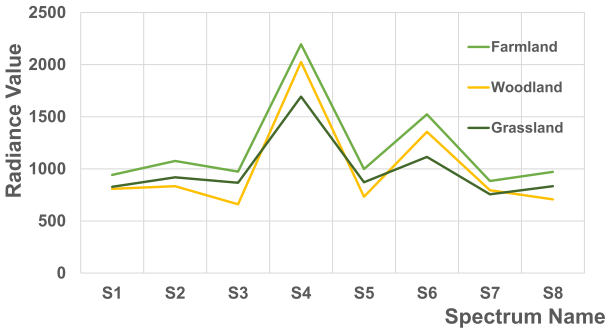



Fig. 1. Radiance value statistics of different vegetation types in different spectrums of GF-6 images.

At present, the representative Earth resources' observation satellites that provide red edge spectrums include RapidEye [15], WorldView-2 [16], and Sentinel-2 [17]. In 2018, the GF-6 satellite was launched, equipped with a 2-m panchromatic camera, an 8-m multispectral high-resolution camera, and a 16-m multispectral medium-resolution wide-view camera. The 16-m camera has an observation width of 800 km, and with the addition of the red edge spectrums, the ability of GF-6 to monitor vegetation has been greatly improved. The specifications of the 16-m camera are given in Table I. The radiance values are counted for every type of vegetation in the dataset used in this article, and the results are shown in Fig. 1. It can be seen that different vegetation types have dissimilar radiance values for each spectrum, based on GF-6 images. GF-6 also reflects the rapid change in vegetation in several near-infrared spectrums (S4, S5, S6), which verifies the occurrence of REE.

### B. Semantic Segmentation Based on Deep Learning

There are many popular frameworks for deep learning semantic segmentation, such as U-net [18] and Deeplab-v3 [19]. Some semantic segmentation networks are widely recognized for their superior performance in certain tasks. High-resolution network (HRNet) [20] used multibranch structure in a whole network, and the resolution of the feature map remained during the forward propagation of each branch. The three branches of HRNet were individual but information was exchanged between them. In addition, the

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TABLE II
SOME NOTATIONS

| Representation | Explanation |
|---|---|
| feature map | A *width* × *height* × *channel*-dimensional tensor. The sizes of *width*, *height*, *channel* are represented by *W*, *H*, *C*, respectively. |
| $conv_{k\times k}$ | The convolution operation with a k × k size kernel, the default value of *k* is 3. |
| *BN* | Batch normalization operation, a basically normalization operation. |
| *ReLU* | Rectified linear unit, used as an activation function. |
| *mp(f,k×k,s)* | Max pooling operation, works on the feature map *f* as a maximum filter with *k*×*k* kernel and *s* stride. |
| *ap(f,k×k,s)* | Average pooling operation, works on the feature map *f* as an average filter with *k*×*k* kernel and *s* stride. |
| *p(f,W×H×C,v)* | Padding operation. Fill the feature map *f* with the value *v* to the shape W×H×C. |
| *up* | Up-sampling operation, use bilinear interpolation to expand the width and height of the feature map to double the original size. |
| *concate* | The concatenate operation. For two feature maps $f_1 \in \mathbb{R}^{W\times H\times C_1}$, $f_2 \in \mathbb{R}^{W\times H\times C_2}$, the operation of *concate* yields $[f_1, f_2] \in \mathbb{R}^{W\times H\times (C_1+C_2)}$ |
| *split* | The reverse operation of *concate*. For a feature map $f \in \mathbb{R}^{W\times H\times (C_1+C_2)}$, the operation is split $f$ into $f_1 \in \mathbb{R}^{W\times H\times C_1}$ and $f_2 \in \mathbb{R}^{W\times H\times C_2}$ |

concept of the feature map is referenced and defined in Table II. The feature map is the tensor extracted from an input image. In general, the feature map has three dimensions, the horizontal and vertical dimensions are defined as *width* and *height*, and the last dimension is defined as *channel*. The exploration of disentangled nonlocal network (DNLNet) [21] on the nonlocal mechanism improved its ability in semantic segmentation tasks. [22] combined graph reasoning and disentangled learning to tackle extreme foreground–background imbalance and large intraclass variance. Pyramid scene parsing network (PSPNet) [23] used pyramid scenario resolution to enhance the network's global scope information extraction capability. Besides, a DCNN predicted the distance map to boost the performance of learning spatial context for semantic segmentation [24]. ResUNet-a used U-Net backbone, residual connections, atrous convolutions, pyramid scene parsing pooling, and multitasking inference to obtain good segmentation performance on the International Society for Photogrammetry and Remote Sensing (ISPRS) 2-D Potsdam dataset [25]. In recent years, ResUNet was proposed to extract roads and achieved good results [26]. [27] used pyramid self-attention network (PISANet) to extract buildings from different RSIs datasets and obtained high overall accuracy (OA).

### C. Attention Mechanism

The AM was proposed in the field of natural language processing [28], [29]. In recent years, AM has also excelled in the field of computer vision. AM is inspired by the way in which humans think. For example, the human visual system tends to observe the object of an image, which assists judgment and ignores irrelevant information [30]. Similarly, in deep learning, the role of AM is to allow the model to dynamically focus on the part of the input that is helpful to the task. AM is now being incorporated into semantic segmentation networks. ResNeSt [31] added channel-wise AM to ResNet, and it may have weights on different channels and enhance message passing on different scales and features. Adaptive pyramid context network (APCNet) [32] adopted an adaptive context module to use global features as guidance in estimating local affinity coefficients, and then calculated a context vector. Global context network (GCNet) [33] overcame the drawback of excessive computation of a nonlocal mechanism by proposing a global context AM. [34] proposed two vision transformer architectures and only involved matrix multiplications that have been optimized in modern deep learning frameworks. [35] modified a standard ResNet toward the design of a vision transformer and discovered several key components that may improve the network performance.

Generally, AM introduces an attention weight for input and uses it to distinguish auxiliary task information from relatively irrelevant information [36]. This attention weight is mainly learned by adding an additional feed-forward neural network structure to the architecture. Structures such as squeeze-and-excitation block (SEBlock) [37], attention gate [38], position attention module (PAM), and channel attention module (CAM) [39] are frequently used in computer vision tasks.

PAM and CAM were proposed by dual attention network (DANet) [39], which modeled the semantic interdependencies in spatial and channel dimensions, respectively. PAM selectively aggregates the characteristics by a weighted sum of features at all the locations. CAM can integrate the relevant features among all the channel mappings, and then optionally emphasize the mappings where interdependencies exist. These two attention modules improve the feature representation of the semantic segmentation network and help obtain more accurate segmentation results.

## III. PROPOSED SEGMENTATION METHODOLOGY

InfAttNet is a deep-learning-based semantic segmentation network for woodland segmentation. The overall network structure is illustrated in Fig. 2. InfAttNet refers to the standard encoder–decoder structure, but has two encoders and one decoder, which means two inputs and one output. The two encoders make the network extract features from the original image specified later and from infrared spectrums, respectively. Apart from the overall structure, InfAttNet uses three AM blocks: an encoder attention block (EAB), position attention block (PAB), and channel attention block (CAB). These three blocks are located in different positions to aggregate woodland features in different ways.

### A. Infrared Attention Network Framework

The three parts of InfAttNet are the main encoder, the subencoder, and the decoder. The RSIs containing infrared spectrums are input to the main encoder, of which several

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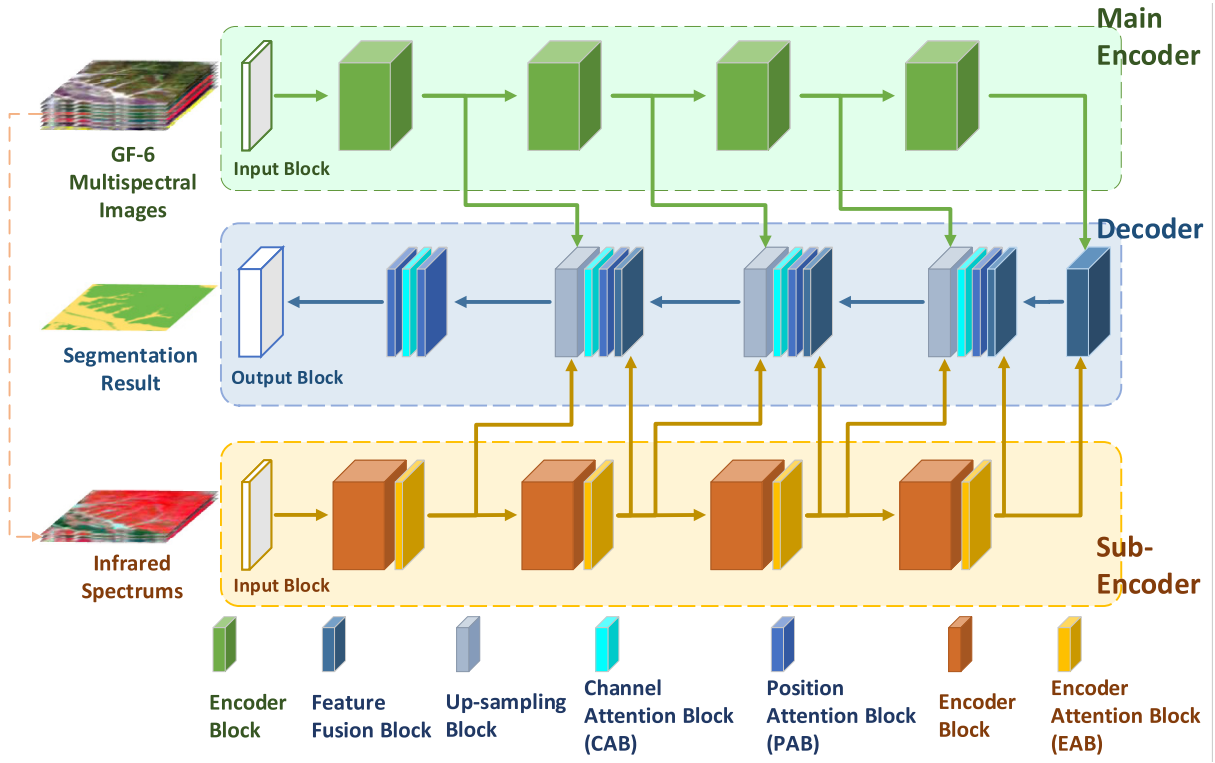



Fig. 2. Overall structure of InfAttNet for woodland segmentation using GF-6 multispectral satellite images. InfAttNet consists of a main encoder, subencoder, and decoder, including three AM modules, i.e., EAB, PAB, and CAB. Note that infrared spectrums contain S4, S5, and S6 as listed in Table I.

specific infrared spectrums are input to the subencoder. After a series of encoding and decoding operations, the segmentation result is output by the decoder.

*1) Main Encoder:* The input to the main encoder is the multispectral image provided by GF-6, which is the original image. The main encoder is composed of one input block and a series of encoder blocks. The input block consists of a classical convolution operation. Its function is to extract common features of the original images. The convolution operation used in InfAttNet is denoted as conv; other operations such as batch normalization (BN) and rectified linear unit (ReLU) are listed in Table II. The input of the block are eight-spectrum RSIs denoted as Img, while the output is a feature map denoted as $x$. The specific operation step is

$$x = \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}(\mathrm{Img}))). \tag{1}$$

The encoder block consists of a downsampling operation and two convolution operations. The series of downsampling operations in InfAttNet enable the network to extract features from an image with different sizes. The max pooling operation, mp, defined in Table II, is used as a downsampling operation. $x_E$ denotes the tensor being operated on the encoder block. The specific operation steps of the encoder block are

$$\begin{aligned} x_{E1} &= \mathrm{mp}(x_E, 2 \times 2, 2) \\ x_{E2} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}(x_{E1}))) \\ x_{E3} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}(x_{E2}))). \end{aligned} \tag{2}$$

These blocks of the main encoder are used to extract common features from multispectral image. They are designed to learn about the characteristics of woodland and other land types. The convolutional operation used in InfAttNet is a general convolutional neural network for feature extraction.

*2) Subencoder:* The input to the subencoder is several infrared spectrums of the original multispectral image. InfAttNet selects S4, S5, and S6 of the RSIs provided by GF-6, because these spectrums contain the red edge wavelength. The REE that is sensitive to vegetation is more obvious in these spectrums, which therefore have more significant value in feature extraction. Section V-D displays experiments specified for spectrum selection of the subencoder input.

The subencoder is composed of an input block, three encoder blocks, and three EABs. The structures of the input block and encoder block are exactly the same as in the main encoder and can be expressed by 1 and 2. In addition to the classic convolution operations and downsampling operations, the subencoder contains an attention module which is called an EAB. This block is introduced in detail in Section III-B and it is mainly used to aggregate the characteristics of vegetation.

*3) Decoder:* After two series of simultaneous encoding operations, the features extracted by two encoders are fully fused in the decoder structure. The InfAttNet decoder consists of upsampling blocks, feature fusion blocks, and an output block. In addition to these blocks, the decoder has two kinds of AM modules: a CAB and a PAB. These attention blocks enable the network to focus on the woodland area in the feature map and facilitate subsequent feature extraction and are introduced in Section III-B.

The upsampling block uses convolution operations with a residual connection and adds an upsampling operation.

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When the channel number of features increases, the residual connection prevents the previous features from being lost [40]. The upsampling operations in the decoder gradually restore the features to the size of the original image. The upsampling operation, up, is defined in Table II. The feature tensor being manipulated in the upsampling block is denoted as $x_U$, and the upsampling block is expressed as

$$\begin{aligned} x_{U1} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}(x_U))) \\ x_{U2} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}(x_{U1}))) \\ x_{U3} &= x_{U2} + x \\ x_{U4} &= \mathrm{up}(x_{U3}, 2 \times 2, 2). \end{aligned} \tag{3}$$

The function of the feature fusion block is to connect the features extracted from the two encoders. It has several residual and convolution structures to supplement the details of the features at this size. The inputs of the feature fusion block contain: $x_{\mathrm{img}}$, which denotes the feature extracted by the main encoder; $x_{\mathrm{inf}}$, which denotes the infrared spectral feature extracted by the subencoder; and $x_{\mathrm{decode}}$, which is the feature from the previous decoder block. In addition to the above expressions, the combined operation which stacks features along the channel dimension is represented by concate. The tensor generated in the feature fusion process is denoted by $x_F$. Note that generally channel is the last dimension of the feature map. Afterward, the operations of the feature fusion block are expressed as

$$\begin{aligned} x_F &= \mathrm{concate}\left(x_{\mathrm{img}}, x_{\mathrm{inf}}\right) \\ x_{F1} &= \mathrm{concate}(x_F, x_{\mathrm{decode}}) \\ x_{F2} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}(x_{F1}))) \\ x_{F3} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}(x_{F2}))) \\ x_{F4} &= x_{F3} + x_{\mathrm{decode}} \\ x_{F5} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}(x_{F4}))) \\ x_{F6} &= x_{F5} + x_F \\ x_{F7} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}(x_{F6}))). \end{aligned} \tag{4}$$

The output block in InfAttNet uses a general convolution structure to regress the feature map and obtain the segmentation results of two categories: woodland and other land types.

### *B. Attention Mechanism Modules*

In this section, three kinds of AM blocks used in InfAttNet are described in detail. Their function is to focus the network on useful vegetation information through different structures.

*1) Encoder Attention Block (EAB):* The EAB is deployed in the subencoder, and its structure is shown in Fig. 3. The subencoder is an encoder for the infrared spectrums in RSIs, helping InfAttNet to screen spectral dimension information, especially for the red edge spectrums. On this basis, the EAB is designed to filter spatial dimension and channel dimension information, which allows the features obtained by the subencoder to concentrate on the vegetation texture and spectral performance.

The structure of the EAB is complex in AMs. As listed in Table II, feature map has three dimensions, called width, height, and channel, respectively. The EAB mainly extracts

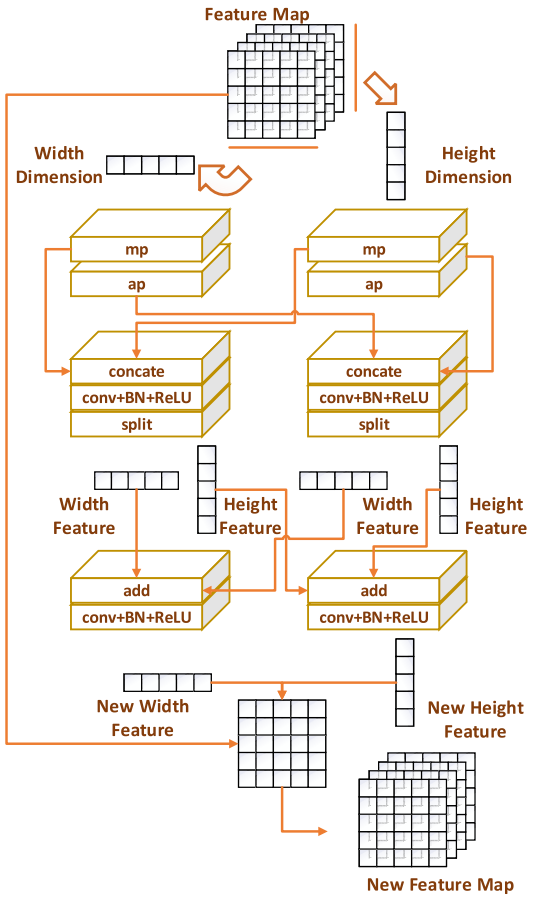



Fig. 3. Structure of EAB.

and aggregates the features from width dimension and height dimension. The EAB first performs mp and ap operations on the width and height dimensions, respectively. The statistics results, which are obtained by different pooling operations, express the information of width and height dimensions. These operations are used as global pooling operations which are referred to the SE-Block [37]. In addition to extracting the average value feature, the EAB also extracts the max value information. Max pooling can filter out useless information and make the characteristics more distinct, while average pooling is toward overall characteristics, which can prevent the loss of information besides the extremum. Since the characteristics of woodland are not uniform, both the maximum and average characteristics must be retained.

As listed in Table II, $W$ represents the width dimension, $H$ represents the height dimension, and the input of EAB is the feature map x. The operations of ap and mp are expressed as

$$\begin{aligned} \mathrm{AP}_H &= \mathrm{ap}(x, 1 \times H, 1), \quad \mathrm{MP}_H = \mathrm{mp}(x, 1 \times H, 1) \\ \mathrm{AP}_W &= \mathrm{ap}(x, W \times 1, 1), \quad \mathrm{MP}_W = \mathrm{mp}(x, W \times 1, 1). \end{aligned} \tag{5}$$

At this point, $\mathrm{AP}_H$ and $\mathrm{MP}_H$ are the feature maps of shape $H \times C$, while the shape of $\mathrm{AP}_W$ and $\mathrm{MP}_W$ is $W \times C$. Before performing the concatenate operation, if $W$ is different from $H$, the feature maps need to be padded with zeros to make them all have the shape $\max(W, H) \times C$. Padding operations

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can be described as

$$\begin{aligned} \mathrm{AP}'_d &= p(\mathrm{AP}_d, \max(W,H)\times C, 0), \quad d \in (W,H) \\ \mathrm{MP}'_d &= p(\mathrm{MP}_d, \max(W,H)\times C, 0), \quad d \in (W,H). \end{aligned} \tag{6}$$

The maximum and average values of the width and height dimensions are then concatenated along the last dimension of the feature maps. Suppose the concatenated result of the average values is $F_{\mathrm{avg}}$, and the concatenated result of the maximum values is $F_{\max}$. The concatenate operation is defined as given in Table II. The concatenate results are expressed as

$$\begin{aligned} F_{\mathrm{avg}} &= \mathrm{concate}\left(\mathrm{AP}'_H, \mathrm{AP}'_W\right) \\ F_{\max} &= \mathrm{concate}\left(\mathrm{MP}'_H, \mathrm{MP}'_W\right). \end{aligned} \tag{7}$$

Next, $F_{\mathrm{avg}}$ and $F_{\max}$ go through the convolution operation

$$\begin{aligned} F'_{\max} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{Con}v_{1\times1}(F_{\max}))) \\ F'_{\mathrm{avg}} &= \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Con}v_{1\times1}\left(F_{\mathrm{avg}}\right)\right)\right). \end{aligned} \tag{8}$$

Then, $F'_{\max}$ and $F'_{\mathrm{avg}}$ are split into two parts along the last dimension, respectively

$$\begin{aligned} \left\{F_{W,\max}, F_{H,\max}\right\} &= \mathrm{split}\left(F'_{\max}\right) \\ \left\{F_{W,\mathrm{avg}}, F_{H,\mathrm{avg}}\right\} &= \mathrm{split}\left(F'_{\mathrm{avg}}\right). \end{aligned} \tag{9}$$

This step is the reverse operation of 7, which is defined in Table II. $F'_{\max}$ is split into $F_{W,\max}$ and $F_{H,\max}$, while $F'_{\mathrm{avg}}$ is split into $F_{W,\mathrm{avg}}$ and $F_{H,\mathrm{avg}}$. If the values of $W$ and $H$ in 5 are different, the padded locations of the feature map need to be removed after split for subsequent operations.

The two width features, $F_{W,\max}$ and $F_{W,\mathrm{avg}}$, are added and convolved, and two height features undergo the same operations. These operations are expressed as

$$F_W = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Con}v_{1\times1}\left(F_{W,\max} + F_{W,\mathrm{avg}}\right)\right)\right) \tag{10}$$

and $W$ is replaced by $H$, when the height dimension is considered.

Referring to coordinate attention [41], the EAB adds more fusion methods and feature extraction methods so that the block adapts to different characteristics of vegetation.

*2) Position Attention Block (PAB):* The PAB is located before the upsampling block in the decoder. The function of the position attention mechanism (PAM) is to obtain the position hint of vegetation, which comes from the subencoder during encoding. The structure of the PAB is similar to that of the PAM [39], but InfAttNet modifies the PAM input and simplifies its structure.

The structure of the PAB is shown in Fig. 4. First, it uses two convolution operations to extract features. The two kinds of feature tensors with different shapes are then tensor-multiplied, and a softmax calculation can make the product result converge between 0 and 1. This result is the weight of each position for the segmentation. Multiplying the weight with the subencoder feature map, and reshaping it gives the same size tensor as the input. Finally, a residual operation is added on the decoder feature map as the output for the PAB, which extracts position information that is more concerned about vegetation. The key and query information of this AM are extracted from the infrared spectrums and used to detail

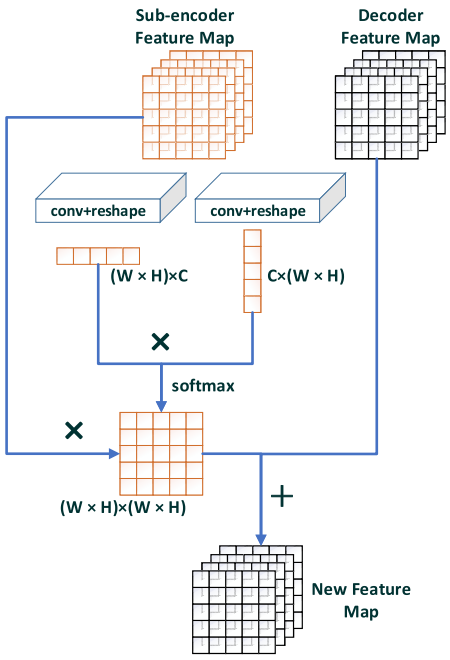



Fig. 4. Structure of PAB. $C$ represents channel number, W represents width, and H represents height. Decoder feature map is the output feature from the previous block, and subencoder feature map is the feature extracted by the subencoder of the corresponding size.

recovery in the decoder. After the information adjustment of the EAB, the features of the subencoder largely aggregate the vegetation characteristics. The positions produced by the subencoder are valuable, and the PAB induces the decoder to pay more attention to them, searching for the effective information under the encoding results. Compared with the PAM, the key and query information extracted by the PAB are based on infrared spectrums, and the extracted feature of the PAB is different.

*3) Channel Attention Block (CAB):* The CAB is located following the feature fusion block in the decoder. The role of this self-attention block is to sort out the most informative channels for woodland segmentation. In fact, the channels of feature map will be doubled quantitatively after feature fusion block, which can cost too many computation resources during decoding without effective dimensionality reduction. This motivates to use the CAB.

The structure of the CAB is illustrated in Fig. 5. Its function is to allow the decoder to find the channels that aid feature extraction. The CAB steps are basically the same as those of the PAB, but the output of the reshape operations and the inputs of blocks are different. First, the CAB performs convolution operations in width and height dimensions. Next, the two outputs are tensor-multiplied by each other. After a softmax operation, the value of key information is put between 0 and 1. The key information and reshaped query information are multiplied to generate an attention tensor. Finally, the fused feature is added to the original feature to accomplish a self-attention work.

After this series of operations, the features are compressed in the channel dimension. Since in the operations each channel

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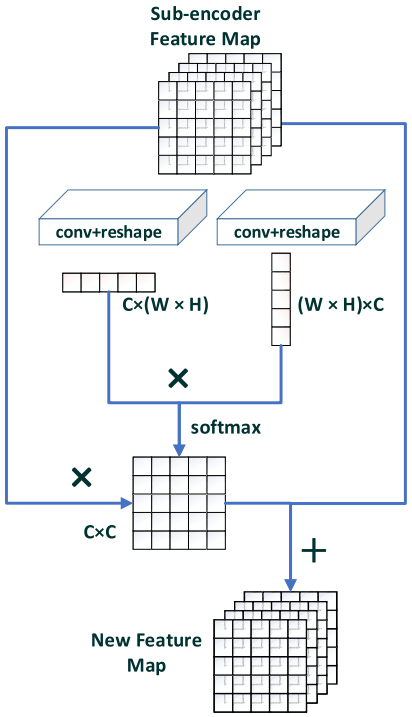



Fig. 5. Structure of CAB. C represents channel number, W represents width, and H represents height. Decoder feature map is the output feature from the previous block.

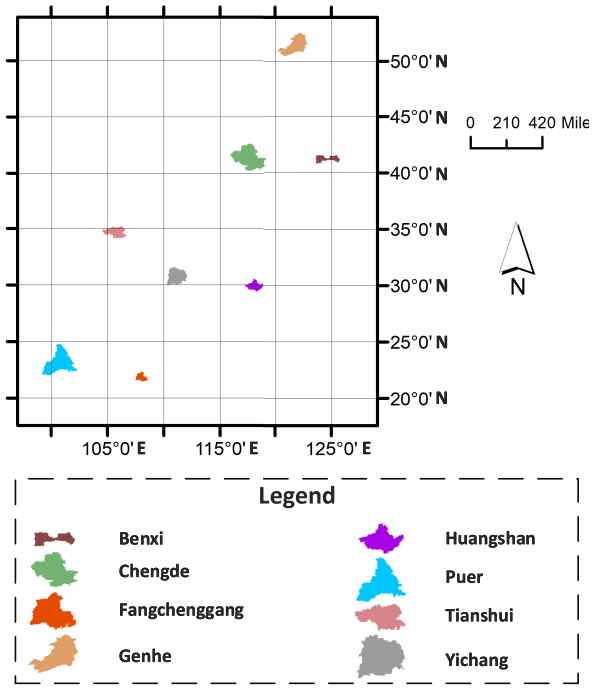



Fig. 6. Distribution of the eight cities selected for the dataset; the location of the city is expressed using latitude and longitude.

gets a value between 0 and 1, representing the weight of the channel. The important channels, i.e., those with large weights, are retained, while the unimportant channels, i.e., those with small weights, are discarded. Due to the adjustment of features, the CAB facilitates subsequent feature extraction.

## IV. Experimental Results and Discussion

### A. Introduction of the Woodland Dataset

The RSIs used in this work are provided by the GF-6 satellite. The spectrum number of the provided RSI is eight; the details are listed in Table I. GF-6 steadily provides RSIs containing infrared spectrums, which are used to prove the validity of InfAttNet. For these reasons, the complete RSIs of eight cities are used as experimental data. Those cities are located in various regions of China, namely, Benxi, Chengde, Fangchenggang, Genhe, Huangshan, Puer, Tianshui, and Yichang. The eight cities cover a total area of more than 163 000 km$^2$. The dataset is distributed over a region spanning approximately 30° of latitude and 25° of longitude and includes a variety of climate types.

The State Forestry and Grassland Administration Survey Planning and Design Institute provided the woodland survey of these areas in 2018. This survey counted the distribution of all the woodlands in China, and the results were mapped by professionals. The survey results were sorted and made into a woodland vector, with marked images based on geographic information science (GIS) technology. GF-6 images corresponding to the years and regions of these cities were searched for and downloaded. The State Forestry and Grassland Administration Survey Planning and Design Institute helped align GF-6 images with survey images. After a radiometric correction operation, these WFV RSIs are projected onto the vector marked images through geographic coordinate information. The corresponding areas are basically congruent, as shown in Fig. 7.

RSIs and matching vector marked images are split, to obtain large numbers of patches of size 256 pixels × 256 pixels. Samples that contain clouds, or have misalignments between the RSI and marked images, are discarded through manual screening. After these steps, 28 999 pairs of images are obtained. Each data group includes an RSI and a survey result image. The various tree species in the survey images are combined into the woodland category, and all other land types are classified as background. The survey images can then be converted into a label image. Fig. 8 shows several groups of data which include an RSI and a label image. About 70% of the groups are randomly selected as the training set, about 15% of the groups as the verification set, and the remaining groups as the testing set. Finally, 20 187 groups of training images, 4448 verification images, and 4364 testing images are obtained.

### B. Settings of Experiments

The experiments are carried out on a computer with a GeForce RTX 3080 graphics processing unit (GPU) with 10 GB of RAM. The optimizer function used in the experiments is the stochastic gradient descent (SGD) function [42], and the learning rate is 0.01.

The loss function used in the experiments is a cross-entropy function. The output of the network is a two-channel segmented prediction tensor. The value of each pixel's channel represents the probability that the pixel is woodland or

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TABLE III

DETAILS OF THE DATASET. EIGHT DIFFERENT CITIES ARE SELECTED AS EXPERIMENTAL AREAS, WHICH HAVE A LARGELY DIFFERENT GEOGRAPHICAL DISTRIBUTION AND CLIMATE. THE NUMBER OF IMAGES CORRESPONDING TO EACH CITY IS RECORDED

| City | Province | Climate | Area($km^2$) | Total Num. | Training | Verification | Testing |
|---|---|---|---|---|---|---|---|
| Benxi | Liaoning | Temperate continental monsson | 8,411 | 1605 | 1117 | 253 | 235 |
| Chengde | Hebei | Temperate continental monsson | 39,519 | 8974 | 6290 | 1317 | 1367 |
| Fangchenggang | Guangxi | Subtropical monsoon | 6,238 | 745 | 503 | 139 | 103 |
| Genhe | Neimenggu | Subfrigid coniferous monsoon | 20,010 | 1807 | 1221 | 297 | 289 |
| Huangshan | Anhui | Subtropical monsoon | 9,807 | 385 | 257 | 61 | 67 |
| Puer | Yunnan | Subtropical monsoon | 44,265 | 8324 | 5769 | 1313 | 1242 |
| Tianshui | Gansu | Temperate continental sub-humid | 14,392 | 2877 | 2029 | 426 | 422 |
| Yichang | Hubei | Subtropical monsoon humid | 21,084 | 4282 | 3001 | 642 | 639 |
| | | | | 28999 | 20187 | 4448 | 4364 |

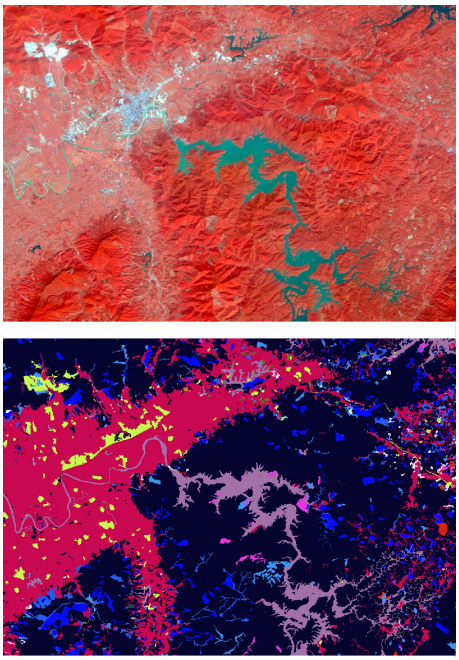


Fig. 7. Above image is a false color image of GF-6, and the under image is the example of the proposed survey result. The region photographed is from Fangchenggang. The different tree species and other land types in the survey result image are indicated by different colors. In the subsequent data processing operations, all tree species are combined as woodland category.

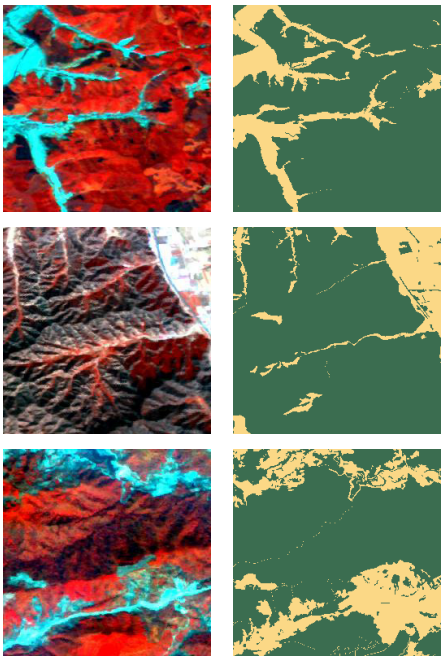


Fig. 8. Left column is the split GF-6 false color images, while the right column is the corresponding label images. The green pixels in label images are denoted as woodland and the yellow pixels are background.

background. Similarly, the label image is expressed as a two-channel tensor of certain events. Suppose the number of image samples is $I$ and the number of classification categories is $N$. In this article, the value of $N$ is 2, which means woodland and background. Assume that the true category of sample $i$ is $y_{i,n}$, and take 1 when $i$ is woodland and 0 when it is background. The predicted probability that sample $i$ belongs to category $n$ is $p_{i,n}$. The loss function of InfAttNet is

$$\text{Loss} = -\frac{1}{I}\sum_{i=1}^{I}\sum_{n=1}^{N} y_{i,n}\log(p_{i,n}). \tag{11}$$

The training epoch is set to 100. Due to GPU's memory limitations, the batch size number is set to 6.

HRNet [20], DANet [39], DNLNet [21], Criss Cross Network (CCNet) [43], GCNet [33], APCNet [32], Deeplab-v3 [19], PSPNet [23], and ResNeSt [31] are selected for comparison with InfAttNet. These frameworks are all deep learning networks with good segmentation results in recent years. Accordingly, several deep neural networks, such as Dense Fully Convolutional Network (DFCN) [7], Convolutional Encoder Neural Networks (CENN) [8], and 2E-UNet [9], are used for comparison. To ensure the accuracy of the experiment, an open-source semantic segmentation toolbox based on PyTorch called MMSegmentation [44] is used to complete the comparison experiments, except for InfAttNet. To ensure fairness, the networks have no preloaded training parameters, and the batch size number is set to 6. The experimental results of each network are calculated by averaging five times. InfAttNet and 2E-UNet have dual encoders, and both of them have eight-spectrum images as inputs in their main encoders, and S4, S5, and S6 of the image in the subencoder of InfAttNet, while 2E-UNet has S4 in its subencoder. All the other network structures have only one encoder, and therefore, they all have eight-spectrum images as inputs.

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TABLE IV
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT SEGMENTATION NETWORKS

| Network | Backbone | OA | AA | MIoU | Woodland | | | Background | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Acc | IoU | F1-score | Acc | IoU | F1-score |
| HRNet [20] | / | 90.3112 | 88.9889 | 79.8978 | 92.3697 | 72.8996 | 92.9886 | 85.6081 | 86.8960 | 84.3259 |
| DANet [39] | ResNet50 | 90.6015 | 89.3224 | 80.4385 | 92.5981 | 87.2608 | 93.1971 | 86.0468 | 73.6163 | 84.8034 |
| DNLNet [21] | ResNet50 | 90.2528 | 89.0562 | 79.7214 | 92.0442 | 86.8639 | 92.9702 | 86.0681 | 72.5788 | 84.1109 |
| CCNet [43] | ResNet50 | 90.2411 | 89.0547 | 79.6934 | 92.0108 | 86.8527 | 92.9638 | 86.0986 | 72.5341 | 84.0809 |
| GCNet [33] | ResNet50 | 90.1249 | 89.0028 | 79.4371 | 91.7601 | 86.7318 | 92.8945 | 86.2456 | 72.1424 | 83.8171 |
| APCNet [32] | ResNet50 | 89.7374 | 88.4804 | 78.7567 | 91.5986 | 86.2292 | 92.6054 | 85.3623 | 71.2842 | 83.2350 |
| Deeplab-v3 [19] | ResNet50 | 89.9335 | 88.5651 | 79.1934 | 92.0495 | 86.4272 | 92.7195 | 85.0806 | 71.9597 | 83.6937 |
| PSPNet [23] | ResNet50 | 85.9975 | 84.0305 | 72.2055 | 88.8925 | 81.6796 | 89.9161 | 79.1685 | 62.7314 | 77.0981 |
| ResNeSt [31] | ResNet101 | 89.1660 | 87.6704 | 77.7966 | 91.4691 | 85.4724 | 92.1672 | 83.8717 | 70.1207 | 82.4364 |
| DFCN [7] | FCN | 87.5881 | 84.7563 | 74.8634 | 92.3564 | 83.6246 | 91.0821 | 77.1561 | 66.1022 | 79.5922 |
| CENN [8] | / | 88.4754 | 85.8572 | 76.4502 | 92.8839 | 84.6892 | 91.7099 | 78.8306 | 68.2112 | 81.1019 |
| 2E-UNet [9] | U-net | 90.2080 | 88.4217 | 79.7725 | 93.2158 | 86.7255 | 92.8909 | 83.6275 | 72.8195 | 84.2723 |
| InfAttNet | ResNet50 | **92.0097** | **90.4966** | **83.1380** | **94.5575** | **89.0371** | **94.2007** | **86.4357** | 77.2390 | **87.1580** |

### *C. Evaluation Metrics*

In this article, OA, average accuracy (AA), and mean intersection over union (MIoU) are selected as evaluation metrics to analyze the overall segmentation results. In addition, Acc (accuracy), IoU (intersection over union), and F1-score are used to evaluate woodland category and background category.

The following indicators are often used to evaluate semantic segmentation methods. True positive (TP) is the pixel number that correctly predicts the target. False positive (FP) is the number of other categories of pixels that predict the target. False negative (FN) is the pixel number that predicts the target incorrectly. True negative (TN) indicates that other pixels are correctly predicted. The formula of Acc is the same as OA, but with a single category as the target. The calculation methods of evaluation indicators are

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN}} \tag{12}$$

$$\mathrm{AA} = \frac{1}{2}\left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}+\frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}\right) \tag{13}$$

$$\mathrm{IOU} = \frac{TP}{\mathrm{FN}+\mathrm{FP}+\mathrm{TP}} \tag{14}$$

$$\mathrm{mIOU} = \frac{1}{k+1}\sum_{i=0}^{k}\frac{\mathrm{TP}}{\mathrm{FN}+\mathrm{FP}+\mathrm{TP}} \tag{15}$$

$$\mathrm{F1\ -\ score} = \frac{2\mathrm{TP}}{2\mathrm{TP}+\mathrm{FN}+\mathrm{FP}}. \tag{16}$$

### *D. Segmentation Performance*

The training set and verification set are used to train InfAttNet and the other contrastive semantic segmentation networks. Since the convolution structure of InfAttNet is mainly based on and modified by ResNet50, the backbone of the selected contrastive networks is ResNet50. ResNeSt with the backbone of ResNet101 is also used for comparisons. Each network is trained five times, and the segmentation scores are averaged to ensure the stability of the model.

Table IV lists the testing results of the segmentation networks. The experimental results demonstrate that the model with the best accuracy is the proposed InfAttNet, which scores 92.01% OA and 83.14% MIoU. HRNet, DANet, DNLNet, CCNet, GCNet, and 2E-UNet have general segmentation results, with OA scores exceeding 90%. DANet is one level below InfAttNet in the results of segmentation and has the second best performance on several metrics. HRNet performs the best in the segmentation of background classes, and its IoU score of background is the highest of all the networks, while its IoU score of woodland is the lowest. In contrast, APCNet, DeepLab-v3, PSPNet, ResNeSt, DFCN, and CENN have worse segmentation results. Their OA scores do not reach 90%, even after being trained multiple times. Among them, PSPNet has the worst segmentation results, with scores of only 86.00% OA and 72.21% MIoU. InfAttNet shows substantial improvement in the accuracy of segmentation. In OA, InfAttNet outperforms DANet by about 1.4%. In AA, it outperforms DANet by about 1.17%. In MIoU, InfAttNet is 2.67% higher than DANet. With the overall evaluation metrics, InfAttNet achieves excellent segmentation results.

In addition, the segmentation accuracy of the woodland category is the metric of the most concern in this task. Table IV indicates that the segmentation results of InfAttNet in the woodland category are also better than the other networks. InfAttNet scores 94.56% on woodland segmentation accuracy, which outperforms DANet by about 1.96%. Similarly, the IoU score of InfAttNet for woodland is 1.78% better than DANet, and its F1-score is 1% better. This indicates that InfAttNet learns the woodland category better than the other networks and can extract the features of woodlands adequately well.

Although OA and AA scores of DFCN, CENN, and 2E-UNet are not high, the segmentation effect for the woodland category exceeds most networks. In particular, 2E-UNet has the second best segmentation on the woodland category after InfAttNet. It is worth stating that 2E-UNet is also a two-encoder network, which proves that this structure is effective for extraction of woodland features. The experiments demonstrate that these three networks have advantages in woodland segmentation, despite the difficulty in extracting diverse features for the background category.

Fig. 9 shows the segmentation results of InfAttNet and several other networks with higher OA on the test set, which contains some typical woodland distribution areas in various experimental areas. Again, InfAttNet has the best segmentation results for the woodland category. The other networks have

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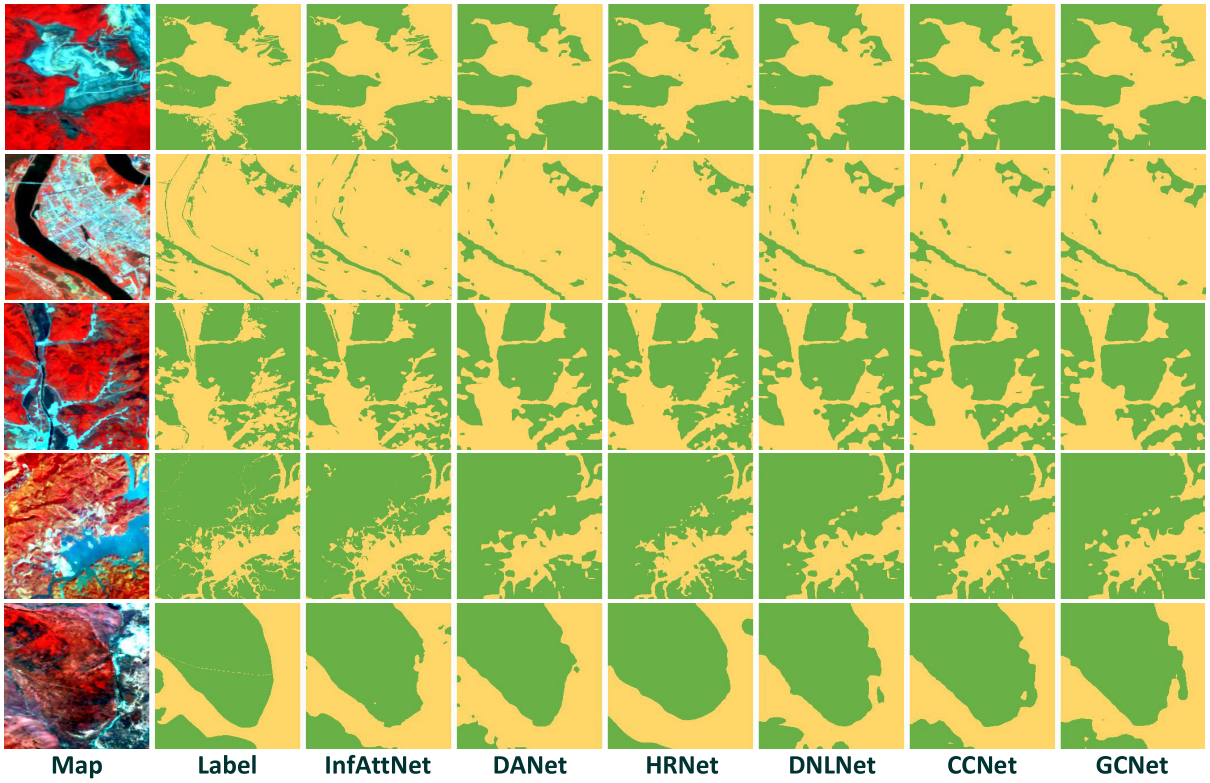



Fig. 9. Segmentation results of InfAttNet and other networks on the test data. It can be seen that the segmentation results of InfAttNet are more prominent in all kinds of details.

different focuses on feature learning for woodland, so the segmentation results differ slightly. HRNet is better in the background learning category, and its segmentation results in handling details are stronger than that of DANet, but not as good as InfAttNet in general. For images with more woodland, the edge treatment of InfAttNet is excellent for detailed processing. For images with less woodland, such as the second row of Fig. 9, InfAttNet also displays better woodland extraction abilities than other networks. In summary, InfAttNet has the best segmentation performance in various areas and for various woodland distribution types. This excellent segmentation is due to our additional encoder with three attention structures.

## V. Ablation Study

To verify the effectiveness of each block, this section progressively adds each structure to the base network and performs ablation experiments. The base network of InfAttNet is U-net, with ResNet50 as the backbone. The results of the experiments are listed in Table V. Note that since some of the PABs and CABs' inputs are features extracted by the subencoder, it is not possible to test the segmentation results of the PAB and CAB without the subencoder.

### A. Ablation for the Subencoder

The experimental results in Table V demonstrate that the subencoder further extracts features of vegetation in the infrared spectrums. Row 1 (R1) and Row 4 (R4) of Table V illustrate that the addition of the subencoder can improve OA from 89.48% to 90.30%, compared with the base network. The experimental results in Section IV indicate that the network has achieved a relatively good segmentation level after the addition of the subencoder.

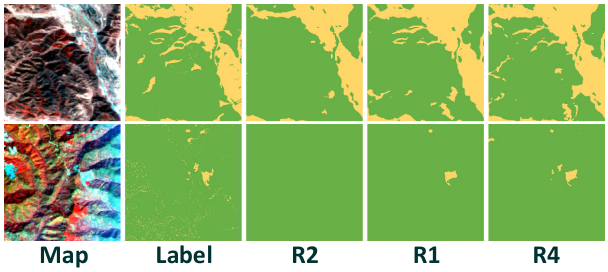



Fig. 10. Results of various ablation experiments on the subencoder.

Apart from that, as listed in Table V, the subencoder is used as an absolute encoder of the network for testing (R2). Three infrared spectrums (S4, S5, and S6) are fed into the subencoder to form a new network and trained. The test results show an OA of 88.43% and segmentation accuracy in the woodland category of 94.39%. This experimental result demonstrates that using three infrared spectrums that are sensitive to vegetation information achieves a good woodland segmentation result.

As shown in Fig. 10, the segmentation results are increased using a subencoder, compared with the original network. The base network does not extract the semantic information of

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TABLE V
SCORES (%) OF INFATTNET AFTER ABLATION OF EACH STRUCTURE

| | Model | | | | | | | | Woodland | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Main-EN | Sub-EN | EAB | PAB | CAB | OA | mIoU | AA | Acc | IoU | F1-score |
| R1 | o | | | | | 89.48 | 78.44 | 87.56 | 92.70 | 85.81 | 92.36 |
| R2 | | o | | | | 88.43 | 76.00 | 84.90 | 94.39 | 84.85 | 91.80 |
| R3 | | o | o | | | 88.98 | 77.01 | 85.59 | 94.69 | 85.50 | 92.18 |
| R4 | o | o | | | | 90.30 | 80.02 | 88.76 | 92.90 | 86.80 | 92.93 |
| R5 | o | o | o | | | 90.78 | 80.89 | 89.31 | 93.26 | 87.41 | 93.28 |
| R6 | o | o | | o | o | 91.02 | 81.22 | 89.21 | 94.08 | 87.79 | 93.50 |
| R7 | o | o | o | | o | 90.82 | 80.64 | 88.32 | 95.04 | 93.42 | 93.42 |
| R8 | o | o | o | o | | 91.09 | 81.38 | 89.37 | 93.99 | 87.87 | 93.54 |
| R9 | o | o | o | o | o | 92.01 | 90.50 | 83.14 | 94.56 | 89.04 | 94.20 |

woodland sufficiently well, which leads to ordinary recognition of woodland. However, with the subencoder, the network extracts more woodland features by aggregating multiple infrared spectrums and obtaining unique red edge information. In InfAttNet, the woodland features extracted by the two encoders are fed into the decoder, enhancing the transmission of woodland semantic information.

### *B. Ablation for the EAB*

The results of the ablation experiments show that the EAB enables InfAttNet to focus on features in different ways, improving its ability to extract features from different categories. Three sets of experiments are conducted to demonstrate the effectiveness of EAB.

R2 and R3 in Table V demonstrate that for a simple network structure, the EAB enhances the segmentation ability of the network to a certain extent. The OA of the network increases from 88.43% to 88.98% and the AA increases from 84.90% to 85.59% after adding the EAB. This indicates that the EAB not only enhances the segmentation network's ability to recognize woodland categories but also enhances the classification of background categories. R4 and R5 in Table V show that the EAB steadily enhances the segmentation ability of the network when its structure is more complex. With the addition of EAB, the network's OA and AA improve. R6 and R9 in Table V indicate that the network accuracy decreases with the removal of EAB, compared with InfAttNet. Although the decrease in the accuracy in the woodland category is smaller, there is a significant decrease in OA.

The function of EAB is consistent with the design purpose, which is to enhance the extraction of whole features of the image. This is due to the fact that the EAB improves feature extraction by paying full attention to different feature types in the feature map. Fig. 11 illustrates that the network structure with the addition of the EAB module improves the overall segmentation results.

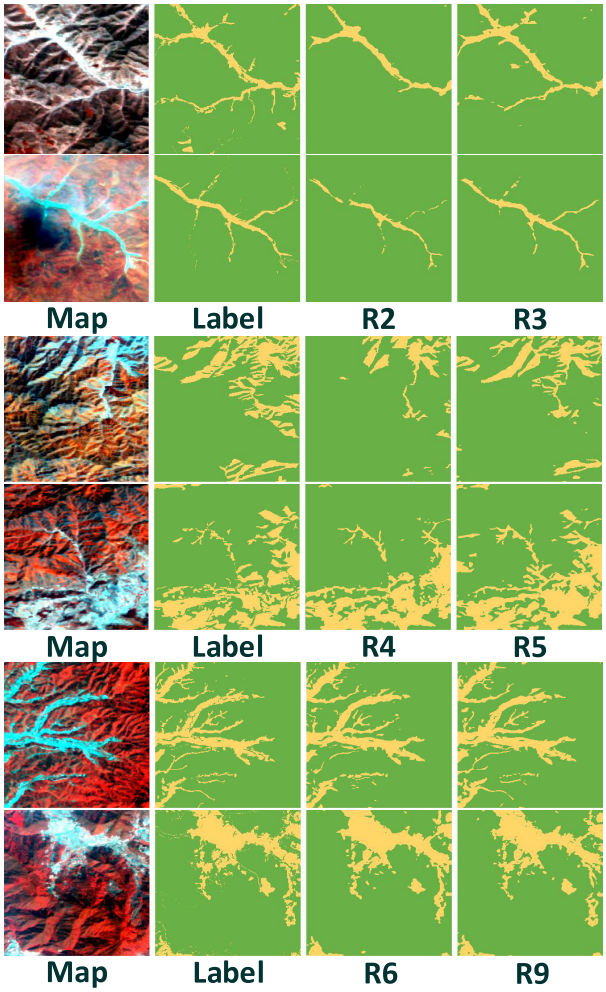



Fig. 11. Results of various ablation experiments on EAB.

### *C. Ablation for the PAB and CAB*

R5 and R8 in Table V show that the addition of PAB enhances the segmentation capability of the network. OA, AA, and woodland accuracy are all improved by adding PAB. Compared with adding only CAB, the overall segmentation results of the network is enhanced more significantly by adding only PAB. R7 and R9 in Table V show that the network's segmentation ability is significantly reduced after removing PAB. OA decreases from 92.01% to 90.82%, a 1.19% decrease, and AA decreases by 2.18%. Removing PAB has the greatest effect on the segmentation of images, compared with removing another block.

R5 and R7 in Table V reflect that the CAB slightly improves the network's OA and significantly improves its ability to segment the woodland category. After adding the CAB, the network's OA increases from 90.78% to 90.82%, and the

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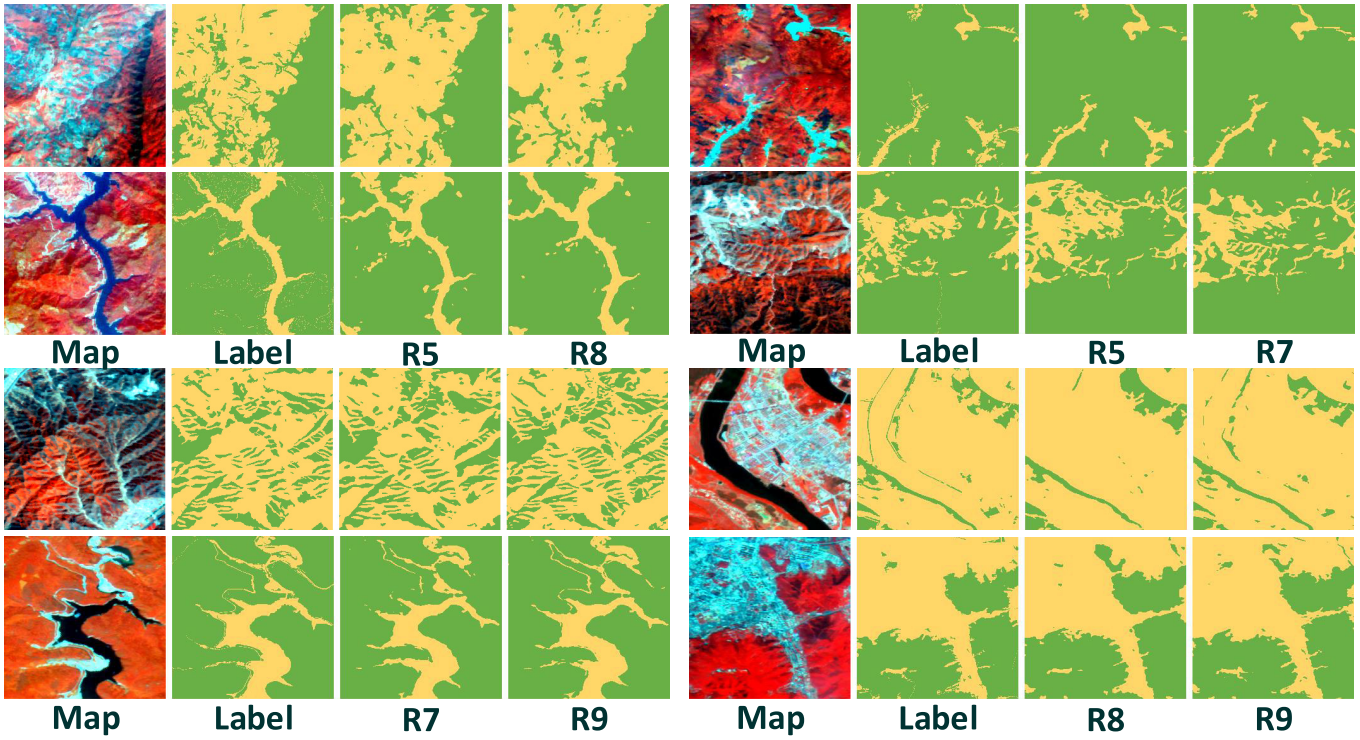



Fig. 12. Results of various ablation experiments on the PAB and CAB.

woodland accuracy increases from 93.26% to 95.04%, which is the highest woodland accuracy so far. In addition, Fig. 12 shows that the segmentation results with CAB are grasped for large areas of woodland during segmentation. This is due to the fact that during the decoding of the network, the CAB performs vector inner product on the multidimensional channel vectors and filters to get more relevant channels. R8 and R9 in Table V confirm that adding CAB can further improve the network's segmentation capability. After adding CAB, OA increases from 91.09% to 92.01% and AA increases from 89.69% to 90.50%. Although this experiment demonstrates a decrease in woodland accuracy, the overall segmentation improves. This is because many images in the proposed dataset have a larger proportion of background category than woodland, as shown in Fig. 12. The network has to learn both the categories sufficiently well if it is to obtain higher OA.

From the above experiments, it can be seen that PAB is the attention module proposed for the distribution of features. Unlike CAB, which focuses on channel information, PAB pays more attention to positional information. Since the distribution of features is relatively critical in semantic segmentation, the proposed PAB which is sensitive to location information improves the segmentation results. The experimental results also demonstrate that PAB is more useful than CAB or EAB for overall segmentation, which confirms that texture information is more important for multispectral semantic segmentation. In Fig. 12, PAB obtains the location information of woodland better and has a good detection performance for small areas.

### *D. Discussion on Input Spectrums of Subencoder*

For RSIs produced by GF-6, S4, S5, and S6 are input to the subencoder of InfAttNet. This is because these infrared spectrums record realistic information about vegetation and preserve the expression of features resulting from the red edge phenomenon. Several sets of experiments are designed to verify this idea. Different combinations of spectrums are fed into the subencoder. The results are listed in Table VI.

The input of S4, S5, and S6 provides the best woodland segmentation performance, confirming that the red edge feature helps the network to analyze woodland features. With additional input spectrums, such as S7 and S8, the correct woodland segmentation rate decreases severely. This is due to the purple and yellow spectrums interfering with the network's extraction of red edge features. It can also be seen from Fig. 1 that S7 and S8 are not very capable of differentiating vegetation.

The segmentation result of woodland by S1, S2, and S4 is also effective and exceeds that of other segmentation networks (R4). This indicates that in addition to REE, the combination of ordinary infrared and other spectrums is also rich in vegetation features. InfAttNet can also be used to extract other types of features, with superior results. The experiment with R5 explores the effect of increasing input spectrums based on R4. The results indicate that simply increasing the input spectrums of the subencoder may lead to a decrease in segmentation capability. In this case, the input of the subencoder increases and the extracted features converge with the main encoder. However, the EAB of the subencoder cannot parse the more important features, leading to decreased segmentation

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TABLE VI
COMPARISON OF THE SCORES (%) AMONG THE DIFFERENT SPECTRUMS CHOSEN

| | Spectrums Combination | OA | IoU | AA | Woodland | | |
|---|---|---|---|---|---|---|---|
| | | | | | Acc | IoU | F1-score |
| R1 | S4\S5\S6 | 92.01 | 83.14 | 90.50 | 94.56 | 89.04 | 94.20 |
| R2 | S4\S5\S6\S7 | 90.61 | 80.56 | 89.01 | 93.31 | 87.22 | 93.17 |
| R3 | S4\S5\S6\S7\S8 | 91.12 | 81.56 | 89.79 | 93.35 | 87.83 | 93.52 |
| R4 | S4\S2\S1 | 91.45 | 82.05 | 89.75 | 94.33 | 88.34 | 93.80 |
| R5 | S1\S2\S3\S4\S5\S6 | 90.76 | 80.79 | 89.07 | 93.59 | 87.42 | 93.29 |
| R6 | S5\S6\S7\S8 | 88.89 | 77.25 | 86.52 | 92.87 | 85.15 | 91.98 |
| R7 | S3\S4\S5\S6 | 90.23 | 79.65 | 87.94 | 94.11 | 86.87 | 92.97 |

capability. The input of S3, S2, S5, and S6 degrades the feature extraction for background category, but the learning of woodland features is excellent and has good segmentation score of woodland category.

The GF-6 satellite has four new spectrums compared with GF-1. These spectrums are fed into the subencoder to explore whether they can improve segmentation performance. After feeding them into the subencoder for training, there is a significant reduction in the segmentation results (R6). After deducing that the red edge features are lost, it is supposed that the features extracted by the extra four spectrums interfere with the feature extraction of the main encoder. It is experimentally proven that S4, S5, and S6 spectrums contain the most conspicuous red edge feature, which is favorable for woodland segmentation. Compared with other spectrum choices, this strategy is more effective.

## VI. CONCLUSION

In this article, a framework called InfAttNet was proposed for woodland segmentation of multispectral RSIs. InfAttNet makes use of the red edge phenomenon of vegetation and identifies woodland from various features of vegetation, through a special network structure. InfAttNet has an additional encoder and several attention blocks with which it extracts the features of infrared spectrums from original images. Therefore, the model is able to preserve vegetation features from multispectral images and used for the identification and segmentation of woodland. The model has been proven to offer superior performance, compared with other advanced methods using semantic segmentation based on deep learning. In addition, a new dataset for woodland segmentation has been collected. This dataset contains GF-6 satellite woodland information for eight cities in China, covering typical woodland patterns in subtropical climates.

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